

Experiment 8

1. Aim

Demonstrate Linux security mechanisms like user authentication, password policies and basic firewall configuration (iptables/ufw).

2. Hardware / Software Required

- **Linux OS** such as Ubuntu, Debian, or Kali Linux
- Terminal
- passwd and useradd utilities
- iptables
- ufw
- libpam-pwquality for password-strength policies (Ubuntu/Debian)

3. Theory

A. User Authentication

Linux uses **user accounts** to control access to the system. Each user has a username, user ID (UID), password and permissions.

Important files include:

File	Purpose
/etc/passwd	Stores basic user account information.
/etc/shadow	Stores password hashes and password-aging information.
/etc/group	Stores group and group-membership information.
/etc/login.defs	Contains default login and password-aging settings.

B. Password Policies

Password policies improve security by requiring users to use strong passwords and, where appropriate, change them periodically. Password complexity can be configured using PAM and pam_pwquality.

A password policy can enforce:

- Minimum password length
- Uppercase and lowercase characters
- Numbers
- Special characters
- Password expiration
- Maximum password age
- Minimum password age

C. Firewall

A firewall controls incoming and outgoing network traffic based on predefined rules.

A firewall monitors and controls network traffic entering or leaving a system. It can **allow trusted connections and block unauthorized or unnecessary connections** based on rules such as IP address, protocol, and port number.

Linux commonly provides:

- **iptables:** A powerful command-line firewall administration tool that works with the Linux netfilter framework. It allows administrators to create detailed rules for controlling network packets.
- **UFW (Uncomplicated Firewall):** A simpler interface for managing firewall rules, particularly suitable for beginners. It allows services and ports to be easily allowed or denied.

For example, SSH traffic normally uses **port 22**, HTTP uses **port 80**, and Telnet uses **port 23**. A firewall can allow required services while blocking insecure or unnecessary ones.

D. Importance of Linux Security Mechanisms

These mechanisms provide **multiple layers of security**. Authentication protects user accounts, password policies strengthen credentials, and firewalls protect the system from unwanted network connections. Together, they help maintain the **confidentiality, integrity, and availability (CIA)** of the Linux system.

4. Procedure

1. Open the Linux terminal and identify the currently logged-in user.
2. Create a new test user account and set a password.
3. Verify the user account and test user authentication.
4. Configure password-aging and password-strength policies for the test user.
5. Check the applied password policy.
6. Configure the UFW firewall with suitable incoming and outgoing rules.
7. Allow required services such as SSH and HTTP while blocking unwanted services.
8. Verify the active firewall rules.
9. Demonstrate basic firewall rules using iptables.
10. Verify the iptables configuration and observe the applied rules.
11. Remove the test user account after completing the experiment.
12. Record the observations and results.

Security Mechanism	Command	Purpose
Check current user	whoami	Displays the username of the currently logged-in user.
Display user details	id	Shows UID, GID and group memberships of the current user.
Create user	sudo adduser testuser	Creates a new user account.
Change user password	sudo passwd testuser	Sets or changes the password of a user.
Switch user	su - testuser	Logs in/switches to another user account.
View user accounts	cat /etc/passwd	Displays basic information about Linux user accounts.
View password information	sudo cat /etc/shadow	Displays password hashes and password-aging information.
Set maximum password age	sudo chage -M 90 testuser	Forces the password to expire after 90 days.
Set minimum password age	sudo chage -m 1 testuser	Prevents the password from being changed again for 1 day.
View password policy	sudo chage -l testuser	Displays password expiration and aging details.
Check UFW status	sudo ufw status	Checks whether the UFW firewall is active.
Enable UFW	sudo ufw enable	Activates the UFW firewall.
Disable UFW	sudo ufw disable	Deactivates the UFW firewall.
Allow SSH	sudo ufw allow 22/tcp	Allows incoming SSH connections.
Block Telnet	sudo ufw deny 23/tcp	Blocks incoming Telnet connections.

View UFW rules	<code>sudo ufw status numbered</code>	Displays configured firewall rules with numbers.
Set default incoming policy	<code>sudo ufw default deny incoming</code>	Blocks incoming connections by default.
Set default outgoing policy	<code>sudo ufw default allow outgoing</code>	Allows outgoing connections by default.
View iptables rules	<code>sudo iptables -L -n -v</code>	Displays current iptables rules and packet statistics.
Allow HTTP using iptables	<code>sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT</code>	Allows incoming TCP traffic on port 80.
Block Telnet using iptables	<code>sudo iptables -A INPUT -p tcp --dport 23 -j DROP</code>	Drops incoming TCP traffic on port 23.

5. Output

```

kali@kali:~$ id
uid=1000(kali) gid=1000(kali) groups=1000(kali),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),4
v,102(scanner),104(bluetooth),113(lpadmin),122(wireshark),123(kaboxer),124(vboxsf)

kali@kali:~$ sudo adduser testuser
warn: The home directory `/home/testuser' already exists. Not touching this directory.
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for testuser
Enter the new value, or press ENTER for the default
  Full Name []: Lunar
    Room Number []: 09
      Work Phone []:
      Home Phone []:
        Other []:
Is the information correct? [Y/n] Y

kali@kali:~$ sudo passwd testuser
New password:
Retype new password:
passwd: password updated successfully

kali@kali:~$ su - testuser
Password:
testuser@kali:~$

```

```

Password:
(testuser@kali)-[~]
$ cat /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
systemd-timesync:x:990:990:systemd Time Synchronization:/:/usr/sbin/nologin
messagebus:x:989:989:System Message Bus:/nonexistent:/usr/sbin/nologin
tss:x:987:987:tss user for tpm2:/:/usr/sbin/nologin
strongswan:x:100:65534::/var/lib/strongswan:/usr/sbin/nologin
tcpdump:x:986:986:tcpdump:/nonexistent:/usr/sbin/nologin
sshd:x:985:65534:sshd user:/run/sshd:/usr/sbin/nologin
_rpc:x:101:65534::/run/rpcbind:/usr/sbin/nologin
statd:x:102:65534::/var/lib/nfs:/usr/sbin/nologin
dnsmasq:x:984:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:103:105:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin
nm-openvpn:x:983:983:NetworkManager OpenVPN:/var/lib/openvpn/chroot:/usr/sbin/nologin
speech-dispatcher:x:982:982:Speech Dispatcher:/run/speech-dispatcher:/bin/false
usbmux:x:104:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
nm-openconnect:x:981:981:NetworkManager OpenConnect plugin:/:/usr/sbin/nologin
pipewire:x:980:980:system user for pipewire:/nonexistent:/usr/sbin/nologin
saned:x:105:106::/var/lib/saned:/usr/sbin/nologin
lightdm:x:106:107:Light Display Manager:/var/lib/lightdm:/bin/false
polkitd:x:979:979:User for polkitd:/:/usr/sbin/nologin
rtkit:x:978:978:RealtimeKit:/proc:/usr/sbin/nologin
colord:x:977:977:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
pcscd:x:976:976:PC/SC Smart Card Daemon:/:/usr/sbin/nologin
geoclue:x:974:974::/var/lib/geoclue:/usr/sbin/nologin
Debian-snmpp:x:107:109::/var/lib/snmpp:/bin/false
ssh:x:108:110::/nonexistent:/usr/sbin/nologin
cups-pk-helper:x:109:113:user for cups-pk-helper service:/nonexistent:/usr/sbin/nologin
redsocks:x:110:114::/var/run/redsocks:/usr/sbin/nologin
_gophish:x:111:116::/var/lib/gophish:/usr/sbin/nologin
iodine:x:112:65534::/run/iodine:/usr/sbin/nologin
miredo:x:113:65534::/var/run/miredo:/usr/sbin/nologin
redis:x:114:117::/var/lib/redis:/usr/sbin/nologin
postgres:x:115:118:PostgreSQL administrator:/var/lib/postgresql:/bin/bash
mosquitto:x:116:119::/var/lib/mosquitto:/usr/sbin/nologin
inetsim:x:117:120::/var/lib/inetsim:/usr/sbin/nologin
mysql:x:973:973:MariaDB Server:/nonexistent:/bin/false
_gvm:x:118:121::/var/lib/ovpnvas:/usr/sbin/nologin
kali:x:1000:1000::/home/kali:/usr/bin/zsh
_dhcpcd:x:972:972:DHCP Client Daemon:/var/lib/dhcpcd:/sbin/nologin
stunnel:x:971:971:stunnel service system account:/var/run/stunnel:/usr/sbin/nologin
uidd:x:975:975:UDD generator helper daemon:/var/lib/libuid:/usr/sbin/nologin
testuser:x:1001:1001:Lunar,09,,:/home/testuser:/bin/bash

```

```

(testuser@kali)-[~]
$

```

```

(testuser@kali)-[~]
$ sudo chage -l testuser
[sudo] password for testuser:
testuser is not in the sudoers file.

(testuser@kali)-[~]
$ exit
logout

(kali@kali)-[~]
$ sudo chage -m 1 testuser

(kali@kali)-[~]
$ sudo chage -m 1 root

(kali@kali)-[~]
$ sudo chage -l root
Last password change          : Mar 20, 2026
Password expires               : Jan 02, 2300
Password inactive              : never
Account expires                : never
Minimum number of days between password change : 1
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7

(kali@kali)-[~]
$ sudo chage -m 1 root

(kali@kali)-[~]
$ sudo chage -l root
Last password change          : Mar 20, 2026
Password expires               : Jan 02, 2300
Password inactive              : never
Account expires                : never
Minimum number of days between password change : 1
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7

(kali@kali)-[~]
$ sudo chage -M 90 root

(kali@kali)-[~]
$ sudo chage -l root
Last password change          : Mar 20, 2026
Password expires               : Jun 18, 2026
Password inactive              : never
Account expires                : never
Minimum number of days between password change : 1
Maximum number of days between password change : 90
Number of days of warning before password expires : 7

```

```
(kali@kali)-[~]
$ sudo cat /etc/shadow
root:*:20532:1:90:7:::
daemon:*:20532:0:99999:7:::
bin:*:20532:0:99999:7:::
sys:*:20532:0:99999:7:::
sync:*:20532:0:99999:7:::
games:*:20532:0:99999:7:::
man:*:20532:0:99999:7:::
lp:*:20532:0:99999:7:::
mail:*:20532:0:99999:7:::
news:*:20532:0:99999:7:::
uucp:*:20532:0:99999:7:::
proxy:*:20532:0:99999:7:::
www-data:*:20532:0:99999:7:::
backup:*:20532:0:99999:7:::
list:*:20532:0:99999:7:::
irc:*:20532:0:99999:7:::
_apt:*:20532:0:99999:7:::
nobody:*:20532:0:99999:7:::
systemd-networkd:*:20532:0:99999:7:::1:
dhcpcd:*:20532:0:99999:7:::1:
systemd-timesyncd:*:20532:0:99999:7:::1:
messagebus:*:20532:0:99999:7:::1:
tss:*:20532:0:99999:7:::1:
strongswan:*:20532:0:99999:7:::1:
tcpdump:*:20532:0:99999:7:::1:
sshd:*:20532:0:99999:7:::1:
_rpc:*:20532:0:99999:7:::1:
statd:*:20532:0:99999:7:::1:
dnsmasq:*:20532:0:99999:7:::1:
avahi:*:20532:0:99999:7:::1:
nm-openvpn:*:20532:0:99999:7:::1:
speech-dispatcher:*:20532:0:99999:7:::1:
usbmuxd:*:20532:0:99999:7:::1:
nm-openconnect:*:20532:0:99999:7:::1:
pipewire:*:20532:0:99999:7:::1:
saned:*:20532:0:99999:7:::1:
lightdm:*:20532:0:99999:7:::1:
polkitd:*:20532:0:99999:7:::1:
rtkit:*:20532:0:99999:7:::1:
colord:*:20532:0:99999:7:::1:
pcscd:*:20532:0:99999:7:::1:
geoclue:*:20532:0:99999:7:::1:
Debian-snmpp:*:20532:0:99999:7:::1:
sshd:*:20532:0:99999:7:::1:
cups-pk-helper:*:20532:0:99999:7:::1:
redsocks:*:20532:0:99999:7:::1:
_gophish:*:20532:0:99999:7:::1:
iodine:*:20532:0:99999:7:::1:
miredo:*:20532:0:99999:7:::1:
redis:*:20532:0:99999:7:::1:
postgres:*:20532:0:99999:7:::1:
mosquitto:*:20532:0:99999:7:::1:
inetsim:*:20532:0:99999:7:::1:
mysql:*:20532:0:99999:7:::1:
_gvm:*:20532:0:99999:7:::1:
kali:$y9T$t40mB0E41xt4E11NwEGix1$uxI/W4.NhSvzduXBHFw8maWHEXXZ1Va4FoT14K4u896:20532:0:99999:7:::
_dhcpcd:*:20630:0:99999:7:::1:
stunnel:*:20630:0:99999:7:::1:
uidd:*:20701:0:99999:7:::1:
testuser:$y9T$t40mB0E41xt4E11NwEGix1$uxI/W4.NhSvzduXBHFw8maWHEXXZ1Va4FoT14K4u896:20702:1:99999:7:::
(kali@kali)-[~]
$
```

```
kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ sudo ufw status
Status: inactive
(kali@kali)-[~]
$ sudo ufw enable
Firewall is active and enabled on system startup
(kali@kali)-[~]
$ sudo ufw allow 22/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)
(kali@kali)-[~]
$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
(kali@kali)-[~]
$ sudo ufw status numbered
Status: active

To Action From
[ 1] 22/tcp ALLOW IN Anywhere
[ 2] 80/tcp ALLOW IN Anywhere
[ 3] 22/tcp (v6) ALLOW IN Anywhere (v6)
[ 4] 80/tcp (v6) ALLOW IN Anywhere (v6)
(kali@kali)-[~]
$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
```

```
kali@kali: ~  
$ sudo iptables -L -n -v  
[sudo] password for kali:  
Chain INPUT (policy DROP 2 packets, 667 bytes)  
pkts bytes target prot opt in out source destination  
6 887 ufw-before-logging-input all -- * * 0.0.0.0/0 0.0.0.0/0  
6 887 ufw-before-input all -- * * 0.0.0.0/0 0.0.0.0/0  
2 667 ufw-after-input all -- * * 0.0.0.0/0 0.0.0.0/0  
2 667 ufw-after-logging-input all -- * * 0.0.0.0/0 0.0.0.0/0  
2 667 ufw-reject-input all -- * * 0.0.0.0/0 0.0.0.0/0  
2 667 ufw-track-input all -- * * 0.0.0.0/0 0.0.0.0/0  
  
Chain FORWARD (policy DROP 0 packets, 0 bytes)  
pkts bytes target prot opt in out source destination  
0 0 ufw-before-logging-forward all -- * * 0.0.0.0/0 0.0.0.0/0  
0 0 ufw-before-forward all -- * * 0.0.0.0/0 0.0.0.0/0  
0 0 ufw-after-forward all -- * * 0.0.0.0/0 0.0.0.0/0  
0 0 ufw-after-logging-forward all -- * * 0.0.0.0/0 0.0.0.0/0  
0 0 ufw-reject-forward all -- * * 0.0.0.0/0 0.0.0.0/0  
0 0 ufw-track-forward all -- * * 0.0.0.0/0 0.0.0.0/0  
  
Chain OUTPUT (policy ACCEPT 3 packets, 120 bytes)  
pkts bytes target prot opt in out source destination  
8 955 ufw-before-logging-output all -- * * 0.0.0.0/0 0.0.0.0/0  
8 955 ufw-before-output all -- * * 0.0.0.0/0 0.0.0.0/0  
4 755 ufw-after-output all -- * * 0.0.0.0/0 0.0.0.0/0  
4 755 ufw-after-logging-output all -- * * 0.0.0.0/0 0.0.0.0/0  
4 755 ufw-reject-output all -- * * 0.0.0.0/0 0.0.0.0/0  
4 755 ufw-track-output all -- * * 0.0.0.0/0 0.0.0.0/0  
  
Chain ufw-after-forward (1 references)  
pkts bytes target prot opt in out source destination  
Chain ufw-after-input (1 references)  
pkts bytes target prot opt in out source destination  
0 0 ufw-skip-to-policy-input udp -- * * 0.0.0.0/0 0.0.0.0/0 udp dpt:137  
0 0 ufw-skip-to-policy-input udp -- * * 0.0.0.0/0 0.0.0.0/0 udp dpt:138  
0 0 ufw-skip-to-policy-input tcp -- * * 0.0.0.0/0 0.0.0.0/0 tcp dpt:139  
0 0 ufw-skip-to-policy-input tcp -- * * 0.0.0.0/0 0.0.0.0/0 tcp dpt:445  
0 0 ufw-skip-to-policy-input udp -- * * 0.0.0.0/0 0.0.0.0/0 udp dpt:67  
0 0 ufw-skip-to-policy-input udp -- * * 0.0.0.0/0 0.0.0.0/0 udp dpt:68  
0 0 ufw-skip-to-policy-input all -- * * 0.0.0.0/0 0.0.0.0/0 ADDRTYPE match dst-type BROADCAST  
  
Chain ufw-after-logging-forward (1 references)  
pkts bytes target prot opt in out source destination  
0 0 LOG all -- * * 0.0.0.0/0 0.0.0.0/0 limit: avg 3/min burst 10 LOG flags 0 level 4 prefix "[UFW BLOCK]"  
  
Chain ufw-after-logging-input (1 references)  
pkts bytes target prot opt in out source destination  
2 667 LOG all -- * * 0.0.0.0/0 0.0.0.0/0 limit: avg 3/min burst 10 LOG flags 0 level 4 prefix "[UFW BLOCK]"  
  
Chain ufw-after-logging-output (1 references)  
pkts bytes target prot opt in out source destination  
  
Chain ufw-after-output (1 references)  
pkts bytes target prot opt in out source destination
```

```
kali@kali: ~  
$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT  
$ sudo iptables -A INPUT -p tcp --dport 23 -j DROP  
$ sudo ufw disable  
Firewall stopped and disabled on system startup  
  
Chain ufw-user-forward (1 references)  
pkts bytes target prot opt in out source destination  
  
Chain ufw-user-input (1 references)  
pkts bytes target prot opt in out source destination  
0 0 ACCEPT tcp -- * * 0.0.0.0/0 0.0.0.0/0 tcp dpt:22  
0 0 ACCEPT tcp -- * * 0.0.0.0/0 0.0.0.0/0 tcp dpt:80  
  
Chain ufw-user-limit (0 references)  
pkts bytes target prot opt in out source destination  
0 0 LOG all -- * * 0.0.0.0/0 0.0.0.0/0 limit: avg 3/min burst 5 LOG flags 0 level 4 prefix "[UFW LIMIT BLOC  
K]"  
0 0 REJECT all -- * * 0.0.0.0/0 0.0.0.0/0 reject-with icmp-port-unreachable  
  
Chain ufw-user-limit-accept (0 references)  
pkts bytes target prot opt in out source destination  
0 0 ACCEPT all -- * * 0.0.0.0/0 0.0.0.0/0  
  
Chain ufw-user-logging-forward (0 references)  
pkts bytes target prot opt in out source destination  
  
Chain ufw-user-logging-input (0 references)  
pkts bytes target prot opt in out source destination  
  
Chain ufw-user-logging-output (0 references)  
pkts bytes target prot opt in out source destination  
  
Chain ufw-user-output (1 references)  
pkts bytes target prot opt in out source destination
```

Conclusion

Linux security mechanisms were successfully demonstrated through user authentication, password policies, and firewall configuration using UFW and iptables. User accounts were authenticated using passwords, and password aging and expiry policies were configured. Firewall rules were implemented and verified to control network traffic, demonstrating essential Linux security and access-control mechanisms.